

# Multi-Speed Field Evaluation of a Semi-Autonomous Precision Sprayer for Weed Control in Vegetable Crops

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**IMPACT:** This study provides the first systematic analysis of speed-accuracy trade-offs in vision-guided precision spraying for vegetable production, a critical gap the prior literature has addressed only fragmentarily. By evaluating a fully integrated YOLOv11 and ByteTrack system across three operating speeds under real field conditions, the work demonstrates that performance degradation originates not from detection alone but from multiplicative losses compounding through the full detection-tracking-actuation pipeline. Rigorous quantification of both weed control and crop damage rates establishes a quantitative empirical baseline for the field. Together, the physically grounded degradation models, five-season training dataset, and prioritized bottleneck analysis and addressing motion-robust detection, spray timing precision, and crop selectivity, provide a concrete development roadmap for next-generation precision sprayers. This work supports the commercialization of site-specific herbicide application in high-value vegetable production by quantifying both current capabilities and the coordinated technical advances required to reach commercial viability.

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## Abstract

Machine vision-guided precision spraying is promising for sustainable vegetable production by enabling site-specific weed targeting, reducing herbicide application while protecting crop yield and quality. However, the effects of operating speed of sprayer systems on the integrated detection–tracking–actuation pipeline remain insufficiently characterized under field conditions. This study evaluated a semi-autonomous, tractor-based precision spraying system for targeted weed control in radish and quantified performance across three operating speeds. A five-season dataset comprising 10,599 images and 84,311 annotated instances across seven classes was curated for model training and evaluation. A YOLOv11-medium model was integrated with ByteTrack tracking and a customized 12-nozzle sprayer on a semi-autonomous electric tractor. Field trials were conducted at 0.13, 0.27, and 0.45 m/s (i.e., 0.3, 0.6, and 1.0 mph) to characterize speed-dependent changes in weed detection, targeting, actuation, and crop damage. At the lowest speed of 0.13 m/s, the system achieved 94.6% mAP@50 and 93.2% recall for weed and crop detection. However, increasing platform speed substantially reduced system performance, with weed hit rate declining from 78.3% to 35.0% and crop contact rates increasing from 20.1% to 28.7%. Consequently, the system did not meet predefined commercial-viability thresholds. These results demonstrate that practical deployment requires improvements beyond weed detection alone, particularly coordinated improvements in motion-robust plant perception, spray-timing compensation, and crop-adjacent actuation precision. The identified performance bottlenecks provide specific engineering targets for advancing vision-guided precision spraying toward practical, high-speed vegetable weed management. The software programs developed in this study have been made publicly available to support future precision weeding research.

**Keywords:** Precision agriculture; Autonomous systems; Smart sprayer; Weed detection; YOLOv11.

## 1. Introduction

Weeds represent one of the most economically damaging threats in global crop production, causing an estimated 34% reduction in attainable yield across major food crops worldwide (Oerke, 2006), with the problem compounded by rapidly growing herbicide resistance now documented in 275 weed species across 76 countries (Heap, 2026). Three decades of stagnation in new herbicide mode-of-action discovery (Westwood et al., 2018) underscores the urgent need for precision weed control strategies that reduce chemical dependency while maintaining production efficiency. Vegetable production systems face particular vulnerability. Weak crop competitive ability, limited registered herbicide options, and critical timing requirements create severe management challenges (Bell et al., 2022). In brassica crops, including radish and cabbage, weed competition during critical periods causes 52–92% yield reductions (Miller and Hopen, 1991; Sutharsan and Srikrishnah, 2015). Recent mechanistic research reveals that weeds reduce yields not merely through resource competition but through early signaling mechanisms involving light quality and volatile compounds, rendering yield loss largely irreversible after narrowing developmental windows (Horvath et al., 2023). Combined with narrow profit margins and limited chemical options, vegetable systems are ideal candidates for precision weeding technologies, achieving site-specific control with minimal crop injury.

Machine-vision-guided precision spraying offers pathways to dramatically reduce herbicide inputs while maintaining efficacy. Field demonstrations document herbicide reductions of 23–89% with cost savings reaching 56% compared to broadcast application (Timmermann et al., 2003; Gerhards, 2022). Deep learning, particularly YOLO-family detectors, has transformed weed detection through robust feature extraction and multi-class recognition across diverse conditions (Chen et al., 2022). Recent research prototypes demonstrate promising capabilities: AI-powered sprayers for tomato and pepper achieved 88–89% precision with <5% missed targets (Vijayakumar et al., 2025), UAV-based spot spraying in maize achieved 94% classification accuracy with 47% herbicide savings at 6 km/h (Allmendinger et al., 2024), and large-scale sugarcane validation demonstrated 97% control efficacy with 35% herbicide reduction (Azghadi et al., 2025). Our own group's prior prototypes illustrate this iterative development: an initial 12-nozzle ATV-mounted sprayer using YOLOv10 achieved a video-based mAP@50 of 48.9% and a 51.3% spraying hit rate under field conditions (Deng et al., 2026a), while a subsequent 24-nozzle, YOLOv11-based upgrade improved field performance to 69.3% mAP@50 and a 74.6% spraying hit rate in a radish field, though crop contact remained substantial (Deng et al., 2026c).

Although substantial progress has been made, current literature lacks a systematic characterization of speed-accuracy trade-offs, specifically varying vehicle speed across controlled levels while concurrently measuring detection accuracy, actuation precision, and weed control efficacy at each speed (Gerhards et al., 2024). A further gap concerns accessibility; most reported systems rely on custom research-grade

hardware, whereas low-cost designs retrofittable to existing tractors, critical for adoption by small and mid-sized vegetable producers, remain comparatively rare (Terra et al., 2021). Previous studies rely on single operational speeds or even static image assessments, neglecting performance degradation caused by motion blur and reduced frame overlap during continuous video processing (Du et al., 2024). Notably, multi-season model robustness remains inadequately characterized, with cross-season testing revealing accuracy degradation up to 14.5% (Deng et al., 2024). Dedicated multi-season data collection efforts remain limited. Recent multi-temporal datasets spanning multiple imaging campaigns have been developed for orchard weed detection (Salcedo-Navarro et al., 2025), but comparable multi-season resources for vegetable row-crop systems, spanning successive growing seasons as pursued in objective (i) of this study, are largely absent from the published literature. Furthermore, in-field crop damage rates are rarely reported specifically for vision-guided precision spraying systems (Gerhards et al., 2024); by contrast, the mechanical weed control literature has documented crop damage extensively, from quantified stand reductions under intelligent cultivators (Lati et al., 2016) to detailed intra-row yield-loss curves spanning a wide range of weed densities (Melander and McCollough, 2021)

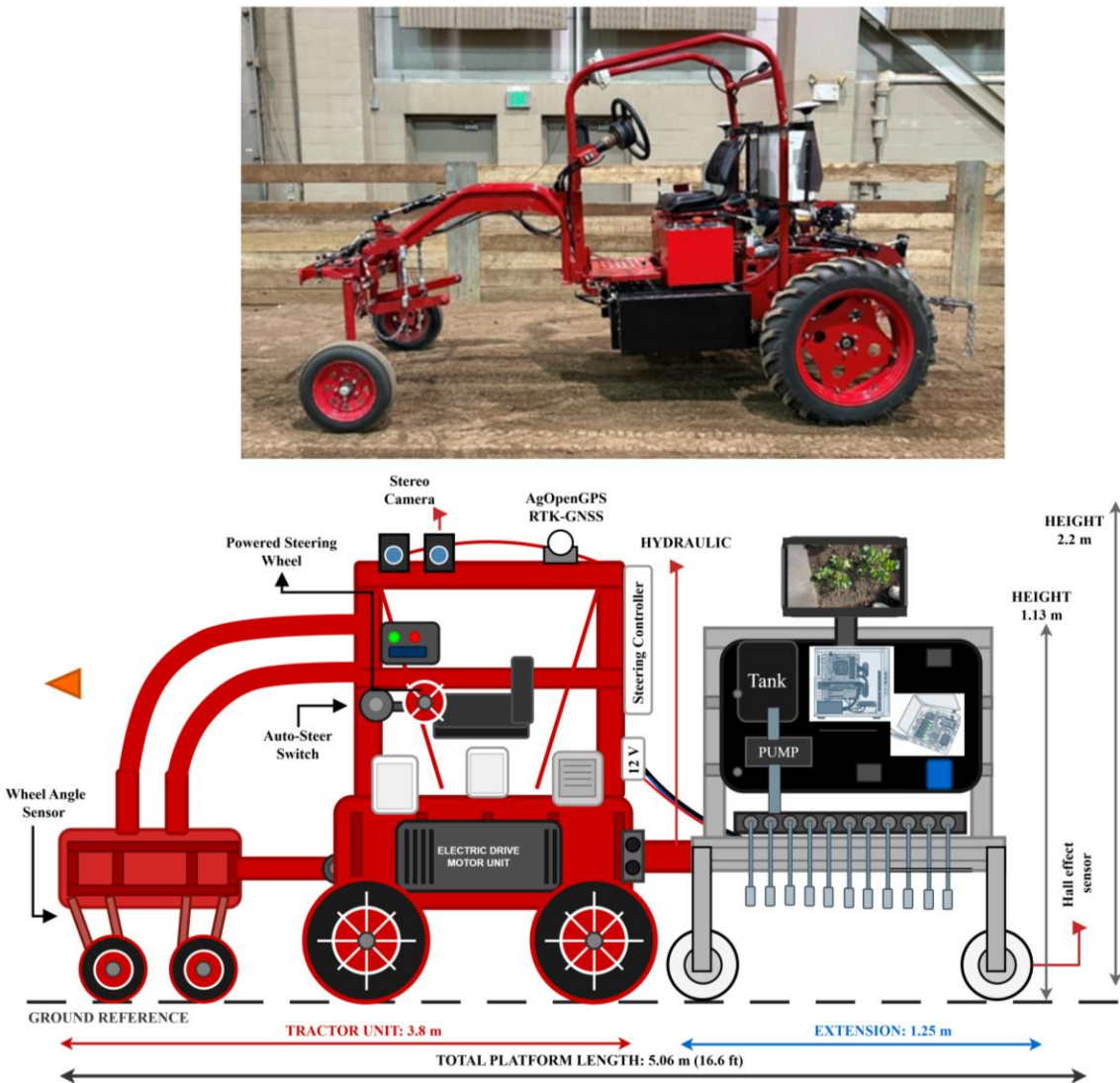
To address these gaps, the study executes a rigorous field-based assessment of a semi-autonomous precision spraying apparatus, prioritizing the quantification of trade-offs between speed and accuracy in production field environments. The methodological framework synergizes a YOLOv11 detection model optimized on a comprehensive five-season dataset (10,599 images, 84,311 instances, 7 target classes), dynamic trajectory tracking via ByteTrack, and a customized 12-nozzle actuation matrix, fully integrated onto an autonomous tractor guided by RTK-GPS technology. Through systematic multi-speed field trials at 0.13, 0.27, and 0.45 m/s (0.3, 0.6, and 1.0 mph), with comprehensive performance assessment, including crop damage quantification. The primary objectives are as follows:

- i. A five-season weed and crop dataset was developed to support robust model training across diverse field conditions, and a YOLOv11-medium detection model integrated with ByteTrack multi-object tracking was trained and evaluated for real-time weed detection on a tractor-based semi-autonomous spraying platform.
- ii. Systematic multi-speed field trials at three operating speeds (0.13, 0.27, and 0.45 m/s) were conducted to characterize speed-dependent degradation across the full detection–tracking–actuation pipeline, including crop damage quantification.
- iii. Key technical limitations, including detection recall degradation under motion, spray-timing latency, and crop-adjacent targeting precision, were identified, and a prioritized improvement roadmap addressing these bottlenecks was established toward achieving commercial viability thresholds.

## 2. Materials and Methods

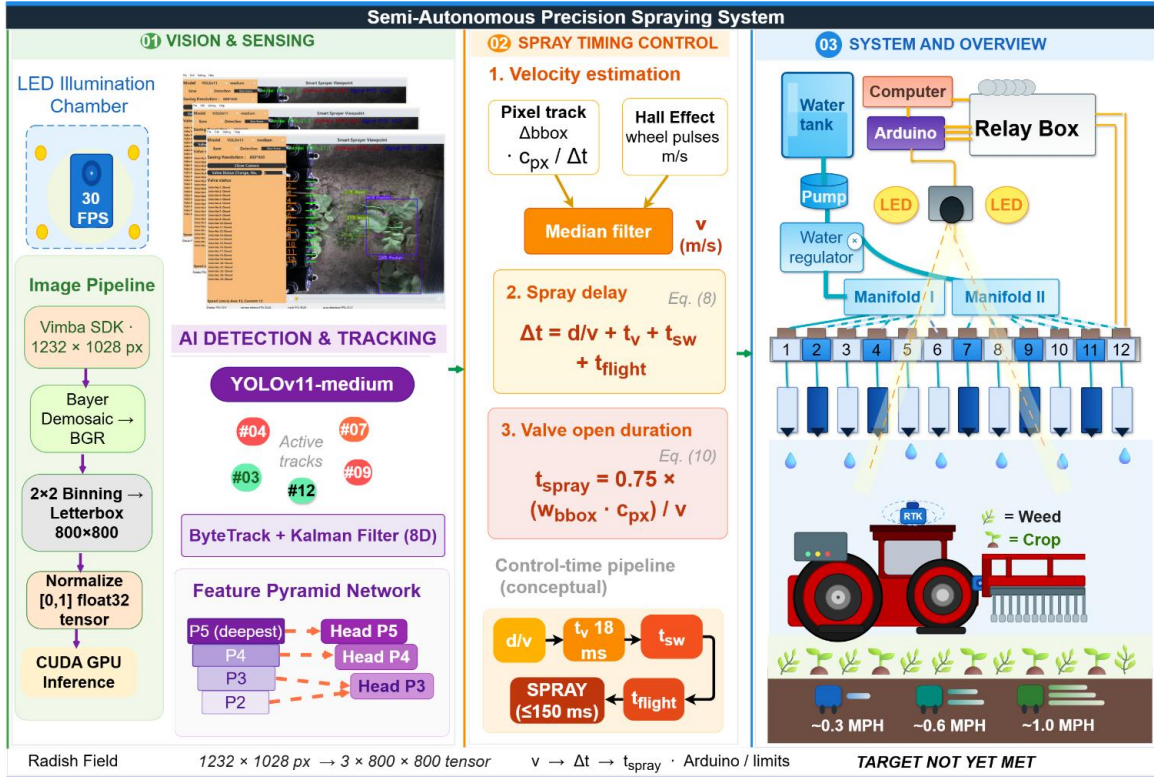
### 2.1 System Architecture and Hardware

**Fig. 1** shows the converted electric tractor platform (§2.2) that served as the base for the precision spraying system described in this section. The precision spraying system integrates YOLOv11-based weed detection, ByteTrack multi-object tracking, and synchronized spray actuation on a semi-autonomous electric tractor platform. The modular architecture employs concurrent processing threads for camera acquisition, detection inference, and valve control, with inter-thread communication via queue-based data structures ensuring real-time coordination without memory accumulation. As shown in **Fig. 2**, the system operates as three coordinated functional subsystems — vision and sensing, spray timing control, and hardware integration — each detailed in the subsections below.



**Fig. 1.** (a) Photograph of the converted electric tractor platform (Oggún II-based; §2.2) used as the base for the semi-autonomous precision spraying system described in this study, and (b) schematic of the semi-autonomous

precision spraying platform integrating navigation, imaging with onboard computing, and a sprayer actuator. The total platform length is 5.06 m (16.6 ft), comprising a 3.81 m tractor unit and 1.25 m spray extension.



**Fig. 2.** System architecture of the semi-autonomous precision spraying platform, organized into three functional subsystems. (1) Vision and sensing: the enclosed imaging chamber acquires frames at 30 frames per second via an industrial camera, which are preprocessed and passed to the YOLOv11-medium detector with ByteTrack multi-object tracking. (2) Spray timing control: platform velocity is estimated from Hall Effect wheel pulses and pixel-based tracking, filtered via a median filter, and used to compute spray delay (Eq. 6) and valve open duration (Eq. 8) for each nozzle. (3) Hardware integration: onboard computer, Arduino relay box, pump, and dual manifolds controlling 12 independently actuated nozzles targeting weeds at three operating speeds (0.3–1.0 mph). The control-time pipeline enforces a  $\leq 150$  ms total response window.

The hardware comprises four integrated subsystems: (1) autonomous navigation platform with RTK-GPS guidance and auto-steering, (2) enclosed imaging chamber with LED illumination and industrial camera, (3) onboard computing unit (Intel i9-11900K, NVIDIA RTX 4060 Ti), and (4) spray actuator assembly with 12 independently controlled nozzles. An extensive configuration system manages hardware calibration, detection thresholds, timing offsets, and filtering parameters, enabling rapid tuning without code modification.

## 2.2 Autonomous Platform and Navigation

The system was deployed on a converted Oggún II tractor (Ronnie Baugh Tractors) (Fig. 1a), a 1,600 lb (726 kg) compact platform assembled from standard off-the-shelf parts, with stock specifications of 150 in. (3.81 m) length, 24 in. ground clearance, independent hydraulic drive, hydraulic steering and brakes, zero-turn capability, a differential lock, and a Category 1 three-point hitch; front track width is adjustable

from 52 to 72 in. in 2 in. increments, with a continuously variable rear track over the same range (Darrow, 2026). For this study, the platform's original 22 HP Honda twin-cylinder gasoline engine was replaced with a 37 HP (27.6 kW) three-phase AC brushless induction motor, a 68% increase in available power over the stock gasoline configuration, drawing from four 88 V (76.8 V nominal) Smartec LiFePO<sub>4</sub> battery packs wired in parallel (56 kg each, ~110 Wh/kg, 24.6 kWh total capacity). The electric motor directly drives a variable-displacement axial-piston traction pump, controlled via a mechanical foot pedal, which supplies hydraulic flow to two fixed-displacement orbital wheel motors on the rear wheels; an auxiliary gear pump mounted on the motor shaft maintains continuous hydraulic flow to the power steering and implement lift cylinders (Darrow, 2026). This platform is distinct from the Farm-Ng Amiga-based prototypes used in our group's prior work (Deng et al., 2026a; Deng et al., 2025c): the Oggún II's larger hydrostatic-drive chassis and three-point hitch provide the load capacity and stability needed for the heavier onboard spray system and extended 1.25 m spray extension used in this study, at the cost of a larger minimum turning footprint relative to the compact Amiga platform. Throttle control remained under direct manual operator supervision, via the foot pedal, to isolate spraying performance from autonomous path planning, while auto-steering maintained precise row following.

Electrical power was distributed across two independent systems. Each Smartec battery pack incorporates an internal DC–DC converter regulated to a 12 V, 15 A output, which supplied the platform's legacy 12 V electrical architecture (key switch, motor management module, CAN-bus termination) as well as auxiliary components including the electric spray pump, solenoid valves, and LED illumination. A separate ECO-WORTHY 12.8 V, 150 Ah LiFePO<sub>4</sub> battery (1.92 kWh nominal capacity), housed in a dedicated enclosure with an integrated 120 A battery management system, provided independent power for the onboard computing unit and camera system, isolating the imaging and detection pipeline from traction-system load transients.

Navigation employed an open-source auto-steering system based on AgOpenGPS (Tischler, 2024) with an on-farm RTK-GPS base station providing differential corrections to a rover-mounted multi-band GNSS receiver via radio link. The auto-steering assembly integrated a powered steering motor, wheel angle sensor, multi-band GNSS receiver, and steering controller, coordinating sensor inputs to generate motor commands. Total navigation retrofit costs approximately \$2,150 (**Supplementary Table S1**), substantially less than commercial systems while achieving comparable performance.

## 2.3 Vision and Computing System

Image acquisition employed an industrial color camera (Alvium G5-508c, Allied Vision Technologies, Stadtroda, Germany) fitted with a 5 mm focal length lens (Kowa, Japan). The camera, which features a 5.1 MP CMOS sensor, was operated with  $2 \times 2$ -pixel binning, giving an effective resolution of  $1,232 \times 1,028$  pixels to balance spatial resolution with processing throughput. The camera was interfaced via Gigabit

Ethernet at 30 FPS, with the exposure time fixed at ~14 ms to prevent motion blur while maximizing brightness under LED illumination. This fixed exposure establishes a fundamental physical constraint on motion-induced image blur. The displacement of a scene element during a single exposure is:

$$\delta_{\text{blur}} = v_{\text{vehicle}} \cdot t_{\text{exposure}} \quad (1)$$

and expressed in pixel units using the calibrated ground-sampling distance  $C_{\text{pixel}} = 0.838$  mm/pixel:

$$\delta_{\text{pixel}} = \frac{v_{\text{vehicle}} \cdot t_{\text{exposure}}}{C_{\text{pixel}}} \quad (2)$$

At the platform forward speed of 0.13 m/s (0.3 mph),  $\delta_{\text{pixel}} \approx 2.2$  pixels; at 0.45 m/s (1.0 mph) this increases to  $\delta_{\text{pixel}} \approx 7.5$  pixels, a 3.4-fold increase that renders sub-centimetre morphological features of small weeds (<2.5 cm) indistinguishable from background texture, providing the physical basis for the speed-dependent recall degradation characterised in Section 3.

The camera was mounted vertically 0.8 m above the ground within an enclosed imaging chamber. Consistent illumination was provided by white LED arrays within the enclosed chamber, which served three critical functions: eliminating sunlight variability and specular reflections, maintaining consistent shadow patterns regardless of ambient conditions, and extending the operating window to low-light conditions. The lightweight aluminum frame with opaque fabric panels excluded ambient light while allowing maintenance access.

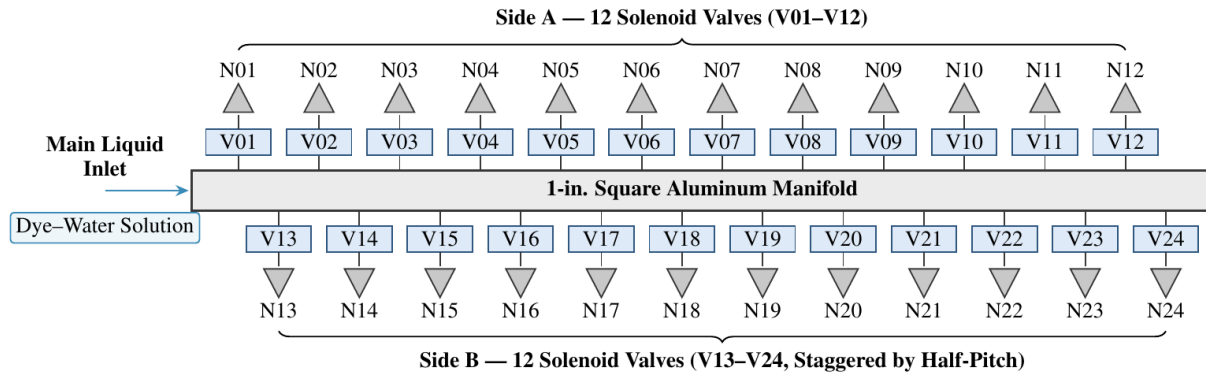
Real-time detection was executed on vehicle-mounted computing (Intel i9-11900K, 64 GB RAM, NVIDIA RTX 4060 Ti GPU with 16 GB VRAM, Ubuntu 22.04, CUDA 12.8). Actual detection throughput ranged from 15–30 FPS, depending on scene complexity. The spatial frame density (detection opportunities per meter of travel) is:

$$\rho_f = \frac{f_{\text{fps}}}{v_{\text{vehicle}}} \quad (3)$$

where  $f_{\text{fps}} = 30$  FPS is the camera frame rate. At 0.13 m/s (0.3 mph),  $\rho_f = 230$  frames/m; at 0.45 m/s (1.0 mph) this falls to 67 frames/m, a 71% reduction in per-plant detection opportunities that directly compounds the blur degradation described in Eqs. 1–2. Complete hardware specifications appear in **Supplementary Table S1**.

## 2.4 Spray Actuator Design

The spray actuator comprised 12 independently controlled nozzles (**Figs. 2–3**) in a linear array at 10 mm center-to-center spacing (total coverage width ~130 mm). This narrow spacing prioritized maximum spatial resolution for targeting accuracy characterization, accepting reduced total coverage as a trade-off.



**Fig. 3.** Schematic (top) and photographs of the nozzle-solenoid valve-manifold assembly in this study.

Each nozzle assembly used modified check valves producing concentrated spray patterns (~15 mm diameter at soil surface). Spray fluid (18.9 L polyethylene tank) was pressurized by a 12 V DC diaphragm pump with pressure regulation at 20 psi, balancing spray pattern formation against drift. Custom manifolds, constructed from 1×1 in. square aluminum tube with 3D-printed end caps sealed using thread-sealing tape and adhesive, positioned nozzles perpendicular to the soil surface. Spray actuation employed normally closed solenoid valves (USS2-00006, U.S. Solid, Cleveland, OH; manufacturer-rated response time  $\leq 1$  s, a conservative full-stroke rating typical of general-purpose valves of this type) controlled via an Arduino microcontroller with automatic safety closure (150 ms timeout) preventing uncontrolled spraying from communication failures. The spray-timing model (§2.6.3) instead assumes a faster ~18 ms nominal response, consistent with the partial-stroke, low-pressure actuation used here rather than the manufacturer's full-stroke rating; this assumption was not independently bench-validated on this hardware (§4.2). Platform velocity was measured via a Hall Effect wheel sensor, supplemented by pixel-based estimation from tracked plant (weed) movement, with median filtering across sources providing robust speed estimation.

## 2.5. Dataset Development

The detection model was trained on a five-season dataset comprising three sub-datasets acquired over five seasons, i.e., 3SeasonWeedDet10, Weed&CropDet2024, and Weed&CropDet2025. The combined dataset contained 10,599 annotated images containing 84,311 object instances across seven target classes (Table 1). This multi-season approach addressed temporal generalization challenges where single-season models exhibit up to 14.5% accuracy degradation in subsequent seasons (Deng et al., 2024).

3SeasonWeedDet10 is a previously published dataset (Deng and Lu, 2025). Weed&CropDet2024 comprises 2024 field weed imagery from Michigan State University, reused here with its original weed-only annotations rather than re-annotated for this study. The Weed&CropDet2025 component (1,161 images, 19,338 instances), collected June–August 2025 using the tractor-mounted platform under deployment conditions, provides training data directly representative of field trial imaging geometry; a related radish precision-sprayer platform and dataset from the same research group are described in Deng et al. (2025), and this study's distinct contribution lies in the multi-speed field characterization enabled by this imagery (§2.7–2.8) rather than in the underlying imagery being wholly exclusive to this work. For field trials, the dataset was consolidated into seven classes reflecting experimental site composition (Table 2).

**Table 1.** Dataset composition across five growing seasons.

| Dataset               | Images        | Instances     | Species/Classes   | Collection Period | Primary Imaging Platform                      |
|-----------------------|---------------|---------------|-------------------|-------------------|-----------------------------------------------|
| 3SeasonWeedDet10*     | 8,436         | 27,963        | 10 weed species   | 2021–2023         | Smartphones, digital cameras, vehicle-mounted |
| Weed&CropDet2024      | 1,002         | 37,010        | 5 crops, 12 weeds | 2024              | Vehicle-mounted platform                      |
| Weed&CropDet2025      | 1,161         | 19,338        | Radish, 6 weeds   | June–August 2025  | Tractor-mounted system (this study)           |
| <b>Combined Total</b> | <b>10,599</b> | <b>84,311</b> | —                 | 2021–2025         | Multiple platforms                            |

\*Previously published (Deng and Lu, 2025).

For the newly collected Weed&CropDet2025 component, bounding box annotations were manually created using LabelImg (Tzutalin, 2015), with each plant instance enclosed by the tightest axis-aligned rectangle and labeled according to the seven target classes (Table 2); the previously published

3SeasonWeedDet10 and Weed&CropDet2024 components retained their original published annotations. All annotations followed YOLO format with normalized center coordinates, width, and height values. Quality control on the newly annotated data employed dual-annotator review of 15% of images with >90% inter-annotator agreement (intersection over union,  $\text{IoU} \geq 0.5$ ). The combined dataset was partitioned into training (75%), validation (5%), and test (20%) subsets using stratified sampling at the field-plot level (preventing data leakage from temporally adjacent frames).

**Table 2.** Target species and classes represented in the combined dataset.

| Class Name      | Scientific Name                     | Role | Notes                                            |
|-----------------|-------------------------------------|------|--------------------------------------------------|
| Radish          | <i>Raphanus sativus</i> L.          | Crop | Primary crop requiring herbicide protection      |
| Carpetweed      | <i>Mollugo verticillata</i> L.      | Weed | Summer annual broadleaf                          |
| Lambsquarters   | <i>Chenopodium album</i> L.         | Weed | Summer annual broadleaf                          |
| Palmer Amaranth | <i>Amaranthus palmeri</i> S. Watson | Weed | Summer annual broadleaf                          |
| Smartweed       | <i>Persicaria</i> spp.              | Weed | Summer annual broadleaf                          |
| Purslane        | <i>Portulaca oleracea</i> L.        | Weed | Summer annual broadleaf                          |
| Weed (general)  | Multiple species                    | Weed | Additional species and uncertain identifications |

Input images were preprocessed to  $800 \times 800$  pixel resolution, preserving aspect ratio, with neutral gray padding (RGB: 128, 128, 128). Data augmentation was applied on-the-fly to training images only (validation and test images were evaluated unaugmented) using the Ultralytics default pipeline: mosaic augmentation, random flips, hue-saturation-value (HSV) perturbations (hue  $\pm 0.015$ , saturation  $\pm 0.7$ , value  $\pm 0.4$ ), scaling ( $\pm 50\%$ ), and translation ( $\pm 10\%$ ).

## 2.6 Detection and Tracking System

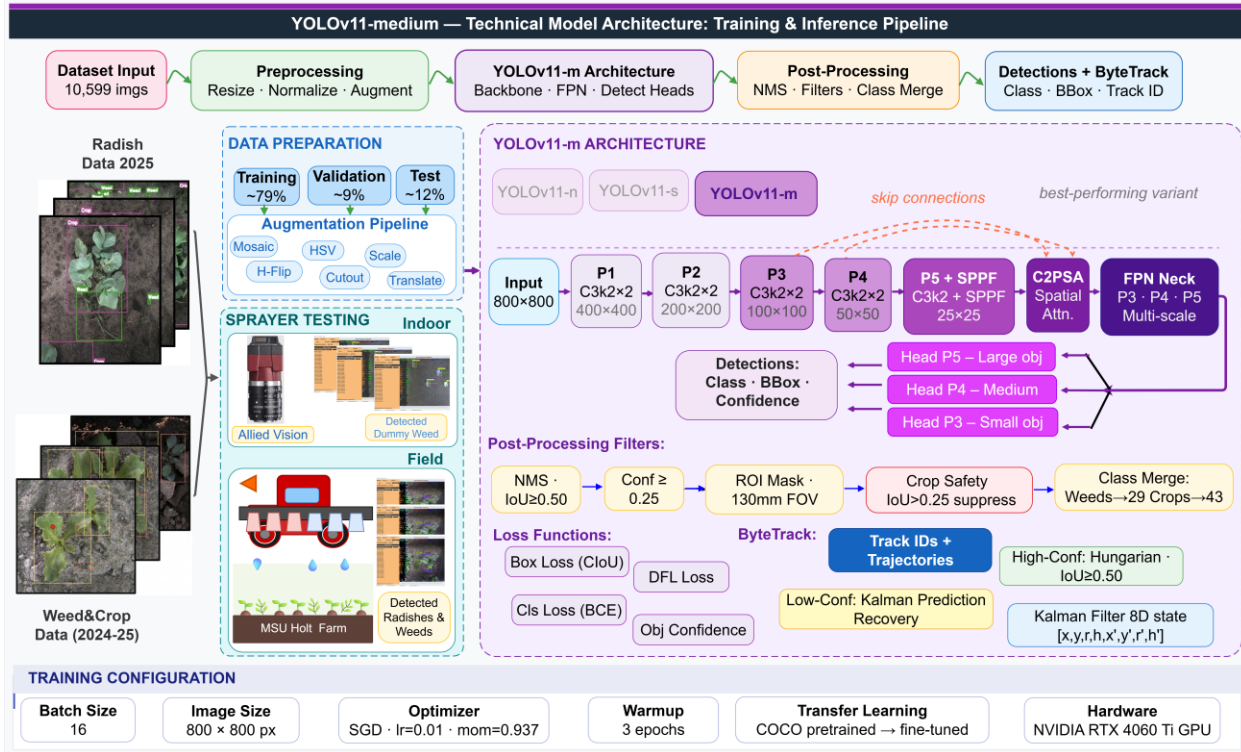
### 2.6.1 Detection Model

Weed detection employed YOLOv11-medium (Jocher et al., 2024), selected for optimal speed-accuracy balance in real-time agricultural applications; the complete training and inference pipeline is summarized in **Fig. 4**. In the broader weed-detection literature, YOLOv11 has achieved 13.5 ms inference with  $\text{mAP}@50$  of 89.0–92.0% across benchmarked weed-detection tasks (Sharma et al., 2024), motivating its selection here for sustained real-time operation without detection latency confounding spray timing. The YOLOv11 architecture incorporates optimized Cross-Stage Partial (C3k2) backbone blocks and spatial attention modules (C2PSA) in the neck, enhancing small object detection critical for early-stage weeds. The detection head generates predictions at three scales, enabling simultaneous detection of small seedlings (<2 cm) and established plants (>10 cm).

Model training utilized the Ultralytics PyTorch implementation (version 11.0.0; Jocher et al., 2024; <https://github.com/ultralytics/ultralytics>), with 36 epochs, a batch size of 8 at  $800 \times 800$  pixel resolution. Optimization employed stochastic gradient descent with momentum ( $\beta = 0.937$ ), initial learning rate  $\alpha_0 = 0.01$  with the standard cosine annealing schedule of Loshchilov and Hutter (2017), adopted without modification:

$$\alpha_t = \alpha_{\min} + \frac{1}{2}(\alpha_0 - \alpha_{\min}) \left( 1 + \cos\left(\frac{t\pi}{T}\right) \right) \quad (4)$$

where  $t$  is the current epoch,  $T = 36$  total epochs, and  $\alpha_{\min} = 0.0001$ . Weight decay  $\lambda = 0.0005$  provided regularization. Complete hyperparameters are summarized in **Supplementary Table S2**. For field deployment, detection thresholds were confidence  $\geq 0.25$  and IoU  $\geq 0.5$  for non-maximum suppression.



**Fig. 4.** Overview of the YOLOv11 model family evaluated in this study, including training pipeline and inference architecture. Three model variants (YOLOv11-nano, small, and medium) were compared on the five-season dataset; YOLOv11-medium was selected for field deployment based on superior  $mAP@50$  (89.9%) while maintaining real-time processing compatibility at 30 FPS. The dataset is first prepared and augmented, then used to train the detector. During inference, the model applies post-processing filters, removes duplicate detections, and tracks objects using ByteTrack. Training was performed with the settings shown (image size, batch size, optimizer) on an NVIDIA GPU.

## 2.6.2 Multi-Object Tracking

Because a given plant's frame-level detection can be inconsistent across consecutive frames under motion blur and partial occlusion, spray triggering cannot rely on single-frame detections alone; multi-object tracking was used to maintain persistent plant identities across frames and supply the stabilized

position and velocity estimates required by a spray-timing model (§2.6.3). Multi-object tracking employed ByteTrack (Zhang et al., 2022), which maintains persistent object identities across frames through two-stage association: high-confidence matching (confidence > 0.25) using IoU-based Hungarian assignment (Bewley et al., 2016), followed by low-confidence recovery (0.1 < confidence < 0.25) reducing track fragmentation from temporary detection failures. State estimation for each track employed a Kalman filter with an eight-dimensional state vector representing center position  $(x, y)$ , aspect ratio  $r$ , height  $h$ , and their respective velocities, following the standard linear constant-velocity motion model:

$$\begin{aligned}\mathbf{x}_t &= \mathbf{F} \mathbf{x}_{t-1} + \mathbf{w}_t \\ \mathbf{z}_t &= \mathbf{H} \mathbf{x}_t + \mathbf{v}_t\end{aligned}\tag{5}$$

where  $\mathbf{F}$  is the  $8 \times 8$  state transition matrix implementing constant-velocity dynamics (position, aspect ratio, and height each updated by their respective velocity terms over the frame interval),  $\mathbf{H}$  is the  $4 \times 8$  observation matrix extracting the directly observed quantities  $(x, y, r, h)$  from the full state vector, and  $\mathbf{w}_t, \mathbf{v}_t$  represent process and measurement noise with empirically tuned covariances (Kalman, 1960).

Kalman filtering has demonstrated robust real-time state estimation across diverse sensing applications, including adaptive multi-algorithm frameworks for structural gap detection using low-cost sensors (Islam et al., 2025). Beyond the standard Kalman-filter update, ByteTrack's association step in this implementation extended the standard IoU-based association cost (cost = 1 - IoU) with a directional soft boost, developed specifically for this system's leftward-drift geometry rather than adapted from prior tracking literature, applied selectively to candidate pairs exhibiting leftward motion, consistent with the apparent leftward drift of stationary weeds through the camera's field of view as the platform advances. For track-detection pairs whose center displacement fell within  $\pm 30^\circ$  of horizontal in the leftward direction, a bonus term combining angular alignment and relative bounding-box area was added to the IoU similarity before rescaling: bonus =  $0.1 \times (-0.5|\theta|/30 + \text{area\_ratio})$ , where  $\theta$  is the displacement angle from horizontal and area\_ratio is the ratio of the smaller to larger bounding-box area between the candidate pair. The adjusted similarity was computed as  $\text{sim}' = \min(1.0, (\text{IoU} + \text{bonus})/1.1)$ , with the corresponding cost taken as  $1 - \text{sim}'$ ; pairs falling outside the leftward cone, including all rightward-moving candidates, retained the unmodified cost of  $1 - \text{IoU}$ . This soft preferential weighting, rather than a hard exclusion of rightward matches, was intended to reduce association ambiguity between differently sized, closely spaced weeds under the platform's characteristic leftward apparent motion; no formal ablation quantifying its effect on track-switch or ID-switch rates was conducted in this study.

### 2.6.3 Spray Timing and Actuation Control

Tracked position and velocity estimates provide the inputs to the spray-timing model: each track's current position, combined with the current vehicle velocity estimate, determines when the corresponding nozzle must fire for the tracked plant to be centered beneath it at the moment of spray release. Precise

spray timing accounts for physical camera-to-nozzle separation, vehicle velocity, and three fixed latency components: solenoid valve response (~18 ms, an assumed nominal value, not independently measured on this hardware), software processing (20.9 ms, measured; **Table 5**), and droplet flight time (~25 ms, also an assumed nominal value), together giving a total fixed delay of ~64 ms; the assumed terms were not bench-validated on the study platform (see **§4.2 Limitations**). The spray trigger time delay  $\Delta t$  for each nozzle is:

$$\Delta t = \frac{d_{\text{camera-nozzle}}}{v_{\text{vehicle}}} + t_{\text{valve}} + t_{\text{software}} + t_{\text{flight}} \quad (6)$$

where  $d_{\text{camera-nozzle}}$  is the longitudinal separation (8–20 cm),  $v_{\text{vehicle}}$  is the current platform velocity (0.13–0.45 m/s),  $t_{\text{valve}}$  is the assumed solenoid response time (~18 ms),  $t_{\text{software}}$  is the measured processing latency (20.9 ms; Table 5), and  $t_{\text{flight}}$  is the assumed droplet travel time (~25 ms), giving a total fixed latency of ~64 ms. This framework demonstrates a speed-dependent positional error:

$$\epsilon_{\text{position}} = v_{\text{vehicle}} \cdot \Delta t_{\text{total}} \quad (7)$$

At 0.13 m/s (0.3 mph), total latency ~64 ms produces  $\epsilon \approx 8.3$  mm. At 0.45 m/s (1.0 mph), identical latency produces  $\epsilon \approx 29$  mm, a 3.5-fold increase consistent with the observed targeting degradation at higher speeds; note that the valve and droplet-flight terms are assumed nominal values rather than bench-measured quantities (see **§4.2**). Valve opening duration was calculated as:

$$t_{\text{spray}} = 0.75 \times \frac{w_{\text{bbox}} \cdot c_{\text{pixel}}}{v_{\text{vehicle}}} \quad (8)$$

where  $w_{\text{bbox}}$  is the maximum bounding box width (pixels),  $c_{\text{pixel}} = 0.838$  mm/pixel, and a factor of 0.75 to prevent over-application; valve duration was bounded between 50 ms and 200 ms.

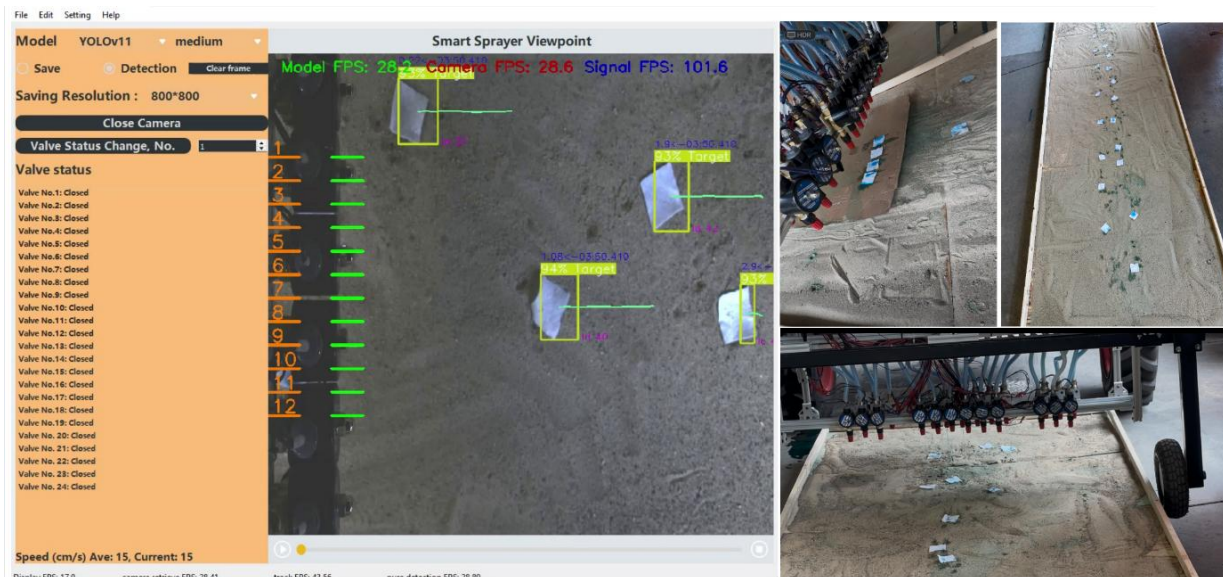
Three post-processing filters were applied to raw detections before spray triggering. The detector's output classes reflect a multi-species label taxonomy retained for cross-season training compatibility; for field spray triggering, binary class merging remapped these predictions to two operational identities — weed (label ID 29) and crop (label ID 43, radish) — since spray decisions depend only on this binary distinction rather than species identity. Region-of-interest filtering discarded detections falling outside the nozzle array's coverage width, since these could not be targeted regardless of detection confidence. Crop overlap filtering suppressed weed detections with  $\text{IoU} > 0.25$  against a detected crop plant, preventing spray triggers for weeds growing in close proximity to crops where actuation risked crop contact.

## 2.7 Experimental Validation

### 2.7.1 Indoor Validation

Initial system validation employed a sand bin test apparatus (0.6 m × 2.4 m) to establish baseline performance under controlled indoor conditions (**Fig. 5**), following a similar validation approach used in

our group's prior sprayer prototypes (Deng et al., 2026a; Deng et al., 2025c), adapted here for the reconfigured 12-nozzle array and updated detection model. One hundred white polyester targets (5 cm × 5 cm) were distributed at ~5 targets/m<sup>2</sup>, simulating field weed spatial arrangement. A simplified color-based detection algorithm (HSV thresholding with morphological operations) replaced YOLO to isolate spray hardware performance from detection variability. The vehicle traversed at 0.27 m/s (0.6 mph, mid-range of field speeds) using blue food-grade dye spray fluid (~5% concentration) for visible deposition marking without phytotoxicity to test targets or, later, field crops. Post-traverse photography was conducted to document hit/miss status and spray positioning.



**Fig. 5.** Indoor validation setup for the precision spraying system. Left: Real-time software interface (Deng et al., 2026a) displaying the YOLOv11 detection output, active nozzle status for all 12 valves, and processing throughput (FPS). Right: Sand bin test apparatus (0.6 m × 2.4 m) with 100 white polyester targets (5 cm × 5 cm) distributed at approximately 5 targets/m<sup>2</sup> to simulate field weed density. Blue dye deposition on targets after each pass was used to quantify weed hit rate and spray positioning accuracy under controlled conditions.

## 2.7.2 Field Trials

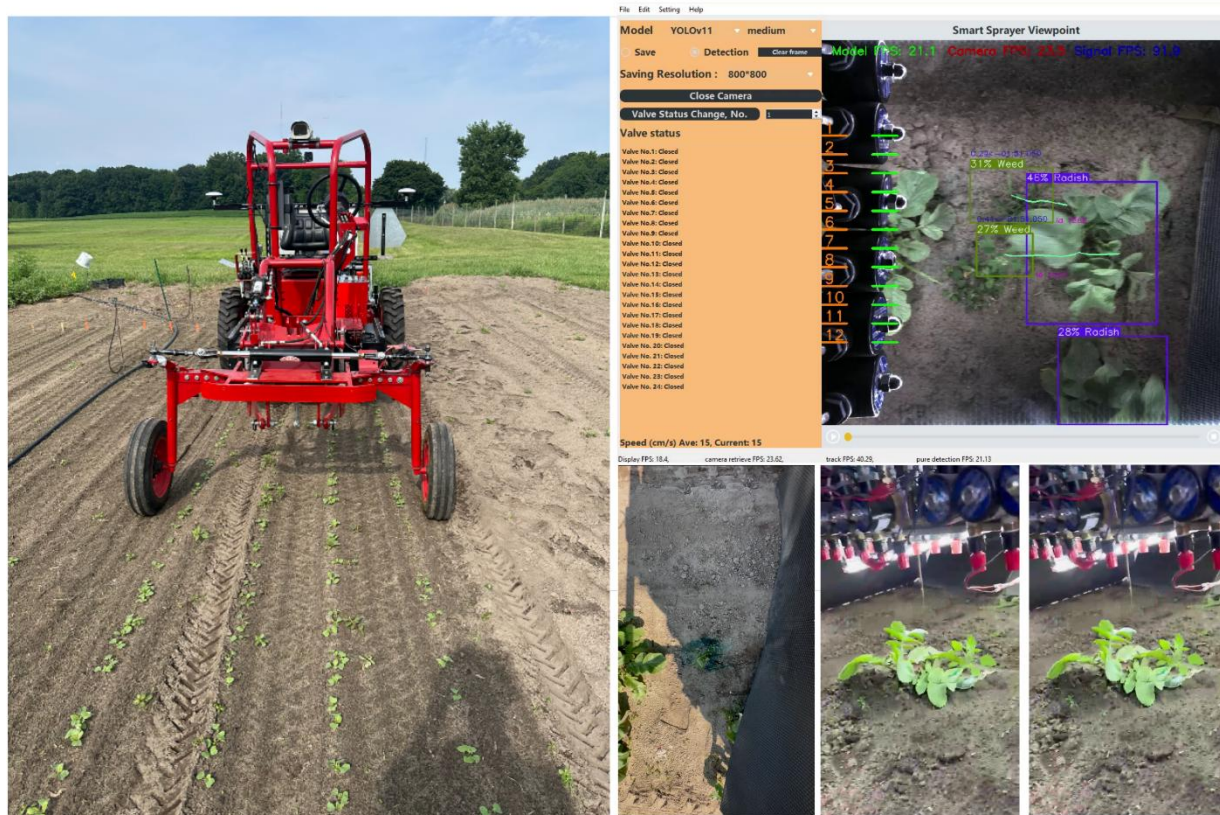
Field experiments were conducted at Michigan State University Horticulture Teaching and Research Center (42.7284°N, 84.4812°W, Holt, MI) on Capac loam soil (pH 6.3, 2.8% organic matter), as shown in **Fig. 6.** Radish (*Raphanus sativus* 'Cherry Belle') was direct-seeded at 2.5 cm in-row × 38 cm between-row spacing in mid-July 2025. At spray trials (mid-August 2025, ~4 weeks post-emergence), radish exhibited 5–7 true leaves with 5–10 cm canopy width, representing the final window for post-emergence herbicide application before canopy closure.

The experimental plot comprised two 30 m parallel rows with the system traversing the inter-row space. Spray coverage width (~13 cm) targeted zones immediately flanking crop rows where weed

competition was most intense. Environmental conditions: 22–28°C air temperature, 55–70% relative humidity, <5 km/h wind, overcast to partly cloudy, moderate soil moisture (~25% volumetric water content), no rainfall within 24 hours. Natural weed populations provided spray targets. Dominant species were common lambsquarters (*Chenopodium album* L.) and common purslane (*Portulaca oleracea* L.), both summer annuals characteristic of Michigan vegetable fields. Weed growth stages ranged from cotyledon to 5+ true leaves (5–25 cm height, 2.5–7.5 cm canopy width) at moderate density (5–10 plants/m<sup>2</sup>). Minimum detectable weed size, as constrained by the imaging and motion-blur characteristics described in §2.3, was 2.5 cm canopy diameter.

#### Field trial experiments

MSU research farm (Holt, MI). Platform in crop rows, YOLOv11 detection, and spray actuation at 0.3, 0.6, and 1.0 mph.



**Fig. 6.** Field trial experiments conducted in August 2025. Top-left: The semi-autonomous spraying platform traversing radish crop rows at the MSU research farm. Top-right: Real-time software interface showing YOLOv11 detection output discriminating between weeds (green boxes) and radish crop (blue boxes) during field operation. Bottom row (left to right): Close-up of inter-row weed populations targeted during trials; blue dye deposition on soil and plant surfaces following a sprayer pass; radish canopy at the 5–7 true-leaf stage present during spray trials.

The platform was operated under direct manual operator control for throttle throughout all trials, with auto-steering (§2.2) maintaining row alignment. Field trials systematically evaluated performance across three speeds, tested in increasing order on the same pair of experimental rows (0.13, then 0.27, then 0.45

m/s; 0.3, 0.6, then 1.0 mph), characterizing degradation from slow research speeds toward the 0.67 m/s (1.5 mph) commercial threshold identified by Vijayakumar et al. (2025), whose reported optimal operating range for tomato and pepper systems (0.45–0.89 m/s) informed the upper end of the tested range. For each speed, the vehicle completed a single continuous pass with YOLOv11 detection and ByteTrack tracking active, using the same dilute blue food-grade dye spray fluid (~5% concentration) described in §2.7.1, with approximately 3–4 h separating consecutive speed treatments; residual dye from earlier passes was not visually distinguishable against soil and plant surfaces after this interval, so contamination of ground-truth scoring for later passes was not expected. Because all three speeds were tested sequentially on the same rows rather than on independent plots, the reported degradation trend reflects a single experimental unit assessed over time rather than replicated spatial units; this design and its implications are discussed as a limitation in §4.2.

## 2.8 Performance Evaluation

### 2.8.1 Detection Performance Metrics

Detection performance was evaluated using three standard object-detection metrics, computed at IoU = 0.5 and confidence  $\geq 0.25$ , including precision, the fraction of detections correctly matching a ground truth object ( $TP / (TP + FP) \times 100\%$ , where TP = true positive and FP = false positive); recall, the fraction of ground truth objects successfully detected ( $TP / (TP + FN) \times 100\%$ , where FN = false negative); and mAP@50, mean Average Precision across N classes at IoU = 0.5 using 101-point COCO interpolation (Lin et al., 2014).

Dataset-based evaluation utilized the held-out test partition (20% of annotated images) representing ideal imaging conditions. Video-based field evaluation extracted frames at ~1 FPS (every 30th frame) from recorded streams, a practical sampling rate chosen to keep manual ground-truth annotation tractable across all three speed treatments rather than one derived from a formal detection-frequency analysis, yielding 70–250 frames per speed treatment. Ground truth annotations for field frames were created using identical protocols as for training data (LabelImg tool, YOLO format). Video-based metrics provide conservative performance estimates, as tracking can recover missed detections through temporal association.

### 2.8.2 Spray Targeting and Crop Protection Metrics

Spray targeting accuracy was quantified through spatial analysis of blue dye deposition relative to ground truth plant locations, using the metrics summarized in **Table 3**. Spray area (SA), weed area (WA), and crop area (CA) were defined by bounding box extents, with centers computed as centroid coordinates. Separate calculations for crop and weed detection enabled discrimination between failure modes affecting targeting versus crop protection. Weed Hit Rate (WHR) serves as the operational proxy for weed control

efficacy throughout this study; the commercial threshold of >90% weed control efficacy (**Table 4**) is equivalent to  $WHR > 90\%$ .

**Table 3.** Spray targeting and crop protection metrics used for system evaluation. WHR = Weed Hit Rate; CCR = Crop Contact Rate; WPR/CPR/SPR = Weed/Crop/Soil Precision Rate. Precision rates (WPR, CPR, SPR) may sum >100% as a single spray event can contact multiple surface types simultaneously.

| Metric                                                        | Definition                                                                                                                                                                         |
|---------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Weed Hit Rate</b><br>( <i>WHR</i> )                        | Percentage of ground-truth weeds receiving spray contact ( $IoU > 0$ ).<br>$WHR = (N_w^{hit} / N_w^{gt}) \times 100\%$                                                             |
| <b>Crop Contact Rate</b><br>( <i>CCR</i> )                    | Percentage of ground-truth crops receiving any spray contact.<br>$CCR = (N_c^{hit} / N_c^{gt}) \times 100\%$                                                                       |
| <b>Weed Precision Rate</b><br>( <i>WPR</i> )                  | Percentage of total spray area deposited on weed targets.<br>$WPR = (A_w / A_T) \times 100\%$                                                                                      |
| <b>Crop Precision Rate</b><br>( <i>CPR</i> )                  | Percentage of total spray area deposited on crop plants.<br>$CPR = (A_c / A_T) \times 100\%$                                                                                       |
| <b>Soil Precision Rate</b><br>( <i>SPR</i> )                  | Percentage of total spray area deposited on bare soil.<br>$SPR = (A_b / A_T) \times 100\%$                                                                                         |
| <b>Average Aligned Distance</b><br>( <i>d<sub>avg</sub></i> ) | Mean Euclidean distance between the spray centroid and weed centroid for successfully hit weeds (mm).<br>$d_{avg} = (1 / N) \sum_{i=1}^N \ SC_i - WC_i\ $                          |
| <b>Average IoU</b><br>( <i>IoU<sub>avg</sub></i> )            | Mean intersection-over-union between the spray area and weed area for successfully hit weeds.<br>$IoU_{avg} = (1 / N) \sum_{i=1}^N (SA_i \cap WA_i) / (SA_i \cup WA_i)$            |
| <b>Translation Efficiency</b><br>( <i>η<sub>trans</sub></i> ) | Ratio of weed hit rate to detection recall; quantifies cumulative loss through the detection–tracking–actuation pipeline.<br>$\eta_{trans}(v) = [WHR(v) / Recall(v)] \times 100\%$ |
| <b>Selectivity Ratio</b><br>( <i>S</i> )                      | Ratio of weed hit rate to crop contact rate; measures discrimination between target and protected plants.<br>$S(v) = WHR(v) / CCR(v)$                                              |

#### Notation

$N_w^{gt}, N_c^{gt}$  total number of ground-truth weed and crop instances, respectively  
 $N_w^{hit}, N_c^{hit}$  number of weed and crop instances, respectively, receiving spray contact  
 $A_T = A_w + A_c + A_b$  total spray area, decomposed into area deposited on weeds, crops, and bare soil  
 $N$  number of successfully hit weeds used in the average-distance and average-IoU calculations  
 $SC_i, WC_i$  spray centroid and weed centroid, respectively, of the  $i$ -th successfully hit weed  
 $SA_i, WA_i$  spray area and weed area, respectively, of the  $i$ -th successfully hit weed  
 $v, Recall(v)$  tractor travel speed and the corresponding object-detection recall at that speed

Commercial viability thresholds were based on agronomic requirements and operational economics (Table 4). These represent concurrent requirements; all three criteria must be satisfied simultaneously, as a deficiency in any dimension renders the system unsuitable for production agriculture.

**Table 4.** Commercial viability thresholds for precision spraying systems.

| Criterion                    | Threshold                 | Rationale                                                                                                                              | Source                           |
|------------------------------|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| <b>Weed Control Efficacy</b> | >90%                      | Conventional herbicides achieve 90–99% control; precision systems must approach this standard to minimize supplemental manual weeding. | Slaughter <i>et al.</i> (2008)   |
| <b>Crop Damage</b>           | <10%                      | Economically acceptable yield-loss threshold for vegetable production with narrow profit margins.                                      | Slaughter <i>et al.</i> (2008)   |
| <b>Operating Speed</b>       | $\geq 0.67$ m/s (1.5 mph) | Minimum for commercial field efficiency in specialty-crop operations; enables timely weed control within critical windows.             | Vijayakumar <i>et al.</i> (2025) |

### 2.8.3 Analytical Approach and Ground-Truth Methodology

Performance analyses addressed three objectives: indoor–field performance gap quantification, speed-dependent degradation characterization, and failure mode categorization. All analyses employed descriptive statistics (means, standard deviations, ranges) as the experimental design focused on comprehensive characterization of a single system across controlled speed treatments rather than hypothesis testing. Given that each speed treatment comprised a single continuous pass on the same experimental rows rather than replicated, independent trials (§2.7.2), inferential statistics were not applicable; characterization was the primary analytical objective.

Indoor–field comparison quantified the performance penalty of transitioning from controlled laboratory to field conditions by computing ratios and absolute differences in detection accuracy, weed hit rate, and targeting metrics at matched speed (0.27 m/s, 0.6 mph).

Speed-dependent analysis characterized degradation trajectories by plotting each metric against operating speed (0.13, 0.27, 0.45 m/s; 0.3, 0.6, 1.0 mph). Linear regression quantified degradation rates (performance change per m/s). The detection-to-spraying performance gap was calculated at each speed to quantify pipeline losses through the detection-tracking-actuation sequence. Given that only three speed treatments were evaluated without replication, regression models represent descriptive trend estimates rather than inferential statistical results.

Failure mode categorization classified unsuccessful spray events (missed weeds, crop contacts) into diagnostic categories: detection failure (weed undetected), tracking failure (track lost before triggering), timing failure (spray displaced >20 mm longitudinally), targeting failure (lateral offset despite correct timing), and intentional crop-proximity suppression ( $\text{IoU} > 0.25$  with crops). Failure mode proportions were

calculated separately per speed to characterize shifting bottlenecks. Complete failure mode taxonomy appears in **Supplementary Table S3**.

Ground truth for spray performance was established through paired pre- and post-spray photography. The forward-facing detection camera described in §2.3 supplied the pre-spray reference frame for each plant, captured as it entered the field of view ahead of the nozzle array. A second RGB camera, mounted at the rear of the platform immediately behind the spray extension, operated continuously at 30 FPS and captured the corresponding post-spray frame as each plant emerged from beneath the nozzle array.

Pre- and post-spray frames were matched by ground location using row-relative distance markers and imported into ImageJ for spray footprint digitization: for each matched pair, the spray deposit was traced to compute its area and centroid, and this centroid was compared against the corresponding plant centroid from the pre-spray frame to determine whether the deposit fell on the intended weed, an adjacent crop plant, or bare soil. Pixel measurements were converted to millimetres using the calibrated ground-sampling distance, yielding the aligned distance and IoU metrics reported in **Table 3**. The resulting hit/miss classification for every detected weed and crop plant in the field trials formed the ground-truth basis for the weed hit rate (WHR) and crop contact rate (CCR) calculations reported in §3.5. To estimate performance at the untested commercial threshold speed, linear regression was fit to each metric (mAP@50, WHR, CCR) as a function of operating speed across the three tested conditions (§3.6).

## **3. Results**

### **3.1 Overview**

Indoor validation at 0.27 m/s (0.6 mph) using simulated targets established baseline capabilities under controlled conditions. Field trials at three operating speeds (0.13, 0.27, and 0.45 m/s; 0.3, 0.6, and 1.0 mph) characterized speed-dependent degradation of detection accuracy, targeting precision, and crop selectivity in radish production with natural weed populations, enabling systematic comparison against commercial viability requirements. §3.2–3.3 report dataset-based and indoor baseline detection performance, addressing objective (i). §3.4–3.5 characterize speed-dependent field degradation across the detection–tracking–actuation pipeline, including crop damage, addressing objective (ii). §3.6 compares field performance against commercial viability thresholds and extrapolates degradation trends to the untested commercial speed, addressing objective (iii). Implications for multi-season generalization and the improvement roadmap are taken up in §4 (**Discussion**).

### **3.2 Dataset-Based Detection Performance**

Three YOLOv11 variants were evaluated on the held-out test partition of the five-season dataset (Table 5). This benchmark was first reported by Deng et al. (2026c); it is reproduced here for completeness and to contextualize the field-deployment results that follow. YOLOv11-medium achieved the highest accuracy

(89.9% mAP@50, 20.9 ms inference) and was selected for field deployment based on superior performance while maintaining real-time processing compatibility at 30 FPS. Overall mAP@50 for nano, small, and medium was 86.2%, 87.2%, and 89.9%, respectively. Beyond the mAP@50 values summarized in Table 5, test-set analysis of the deployed medium model revealed distinct error patterns: weed precision (91.0%) exceeded recall (79.3%), while crop detection showed more balanced performance (88.9% precision, 88.4% recall).

**Table 5.** YOLOv11 variant performance on the five-season dataset. Evaluated on the held-out test partition (20% of the combined five-season dataset). mAP@50 = mean Average Precision at IoU threshold 0.5. Inference time measured on an NVIDIA RTX 4060 Ti GPU.

| Model          | Inference Time (ms) | Crops mAP@50 (%) | Weeds mAP@50 (%) |
|----------------|---------------------|------------------|------------------|
| YOLOv11-nano   | 10.0                | 89.8             | 82.5             |
| YOLOv11-small  | 14.6                | 87.1             | 87.2             |
| YOLOv11-medium | 20.9                | 90.7             | 89.1             |

### 3.3 Indoor Validation Performance

Indoor validation at 0.27 m/s (0.6 mph) demonstrated near-optimal performance (Tables 6 and 7). Detection using the color-based baseline algorithm achieved 99.1% AP@50 (bounding-box level, simulated white targets). Note that indoor AP@50 was computed on simulated white targets using color thresholding; it is not directly comparable to YOLOv11 field mAP@50 and serves only to isolate spray hardware performance from detection variability.

Spraying achieved an 82.0% weed hit rate with a 7.5 mm mean aligned distance and IoU of 0.53. The 18% miss rate despite 99.1% AP@50 demonstrated that spray timing limitations, not detection failures, constrained performance. Missed targets were distributed across nozzle positions without systematic spatial patterns.

### 3.4 Field Detection Performance

Field detection performance degraded progressively with speed (Table 6; Fig. 7). Across 0.13–0.45 m/s (0.3–1.0 mph), mAP@50 declined 24.5 percentage points (94.6% → 70.1%), a 26% relative decrease, with degradation accelerating at higher speeds: the 0.13–0.27 m/s interval produced a 9.7-point decline versus 14.8 points for 0.27–0.45 m/s, confirming non-linear performance loss. Recall degraded more severely than precision (30.7 vs. 19.5 percentage points), with 37.5% of weeds undetected at 0.45 m/s compared to only 6.8% at 0.13 m/s, indicating that weeds became undetectable under motion rather than misclassified. Indoor–field comparison at 0.27 m/s showed a 14.2-point performance reduction (99.1% → 84.9% mAP@50), while the relationship to dataset baseline shifted with speed: at 0.13 m/s, field mAP@50 (94.6%) exceeded the static dataset performance (89.9%), but at 0.45 m/s fell 19.8 points below it (70.1% vs. 89.9%), demonstrating that dataset-based metrics substantially overestimate field capability at higher operating speeds.

**Table 6.** Detection performance across operating speeds. Indoor validation used color-based detection on white targets; field validation used YOLOv11-medium on natural weed and crop populations. mAP@50 = mean Average Precision at IoU threshold 0.5. Speed equivalents: 0.13 m/s = 0.3 mph, 0.27 m/s = 0.6 mph, 0.45 m/s = 1.0 mph.

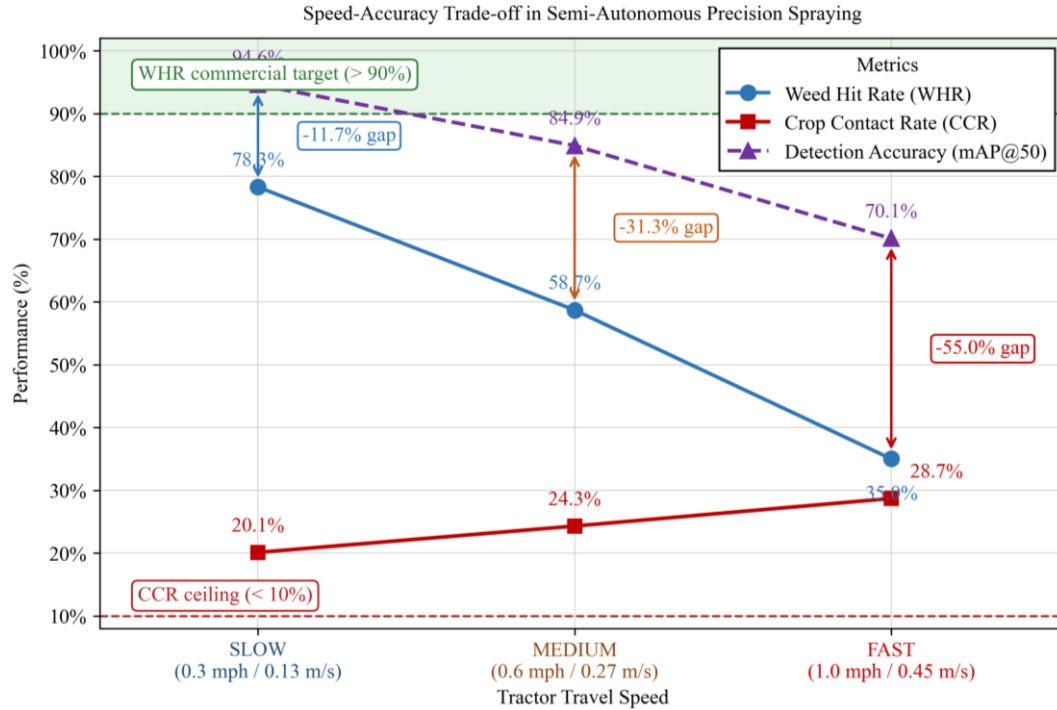
| Test Condition    | Precision (%) | Recall (%) | mAP@50 (%) |
|-------------------|---------------|------------|------------|
| Indoor (0.27 m/s) | 98.1          | 99.5       | 99.1       |
| Field (0.13 m/s)  | 95.8          | 93.2       | 94.6       |
| Field (0.27 m/s)  | 88.4          | 80.7       | 84.9       |
| Field (0.45 m/s)  | 76.3          | 62.5       | 70.1       |

### 3.5 Field Spraying Performance

Weed hit rate declined progressively across the field speed range, from 78.3% at 0.13 m/s (0.3 mph) to 35.0% at 0.45 m/s (1.0 mph) — a 55.3% relative decline within the tested field range, and a 57.3% relative decrease from the 82.0% indoor baseline (**Table 7**). Decline accelerated at higher speeds: a 19.6-point drop between 0.13–0.27 m/s versus 23.7 points between 0.27–0.45 m/s.

Crop contact rates increased with speed (20.1% → 28.7%), exceeding the 10% commercial tolerance at all speeds. Weed:crop selectivity ratios (**Table 7**) declined from 3.9:1 to 1.2:1. Spray deposition increasingly contacted multiple surfaces: at 0.45 m/s (1.0 mph), precision rates summed to 152.6% (71.6% weeds, 38.2% crops, 42.8% soil). **Fig. 7** summarizes these speed–accuracy trade-offs, plotting weed hit rate, crop contact rate, and detection accuracy together against travel speed relative to the commercial viability thresholds established in §2.8. Representative footage of live spray actuation and field-based weed detection and tracking during the multi-speed trials is provided in **Supplementary Video S1**.

Spatial targeting degraded substantially with platform speed. Weed-aligned distance increased 2.1-fold (10.2 → 21.3 mm), while weed IoU declined 50% (0.16 → 0.08). Crop targeting showed similar but less severe degradation (1.5-fold distance increase, 36% IoU decline). The detection-to-spraying performance gap (mAP@50 minus weed hit rate) widened with speed: 16.3 points at 0.13 m/s (0.3 mph), 26.2 points at 0.27 m/s (0.6 mph), and 35.1 points at 0.45 m/s (1.0 mph). Translation efficiency (as defined in **Table 3**) declined from 84% to 56% across the tested speed range (**Table 7**).



**Fig. 7.** Speed–accuracy trade-offs across three operating speeds (0.13, 0.27, and 0.45 m/s) in radish production. Weed Hit Rate (WHR, blue solid), Crop Contact Rate (CCR, red solid), and detection accuracy (mAP@50, purple dashed) are plotted against tractor travel speed. Dashed horizontal lines indicate commercial viability thresholds: >90% WHR (green) and <10% CCR (red). Vertical arrows show the gap between achieved WHR and the commercial target at each speed.

**Table 7.** Spraying performance across operating speeds. Translation efficiency and Selectivity ratio are formally defined in Table 3. Indoor validation used simulated white targets; crop metrics not applicable (—). Precision rates may sum >100% due to spray events contacting multiple surface categories. Aligned distance = Euclidean distance between spray center and target center. IoU = Intersection over Union between spray footprint and target area. Speed equivalents: 0.13 m/s = 0.3 mph, 0.27 m/s = 0.6 mph, 0.45 m/s = 1.0 mph.

| Metric                     | Indoor | 0.13 m/s | 0.27 m/s | 0.45 m/s |
|----------------------------|--------|----------|----------|----------|
| Weed hit rate (%)          | 82.0   | 78.3     | 58.7     | 35.0     |
| Crop contact rate (%)      | —      | 20.1     | 24.3     | 28.7     |
| Translation efficiency (%) | 82.4   | 84.0     | 72.7     | 56.0     |
| Selectivity ratio          | —      | 3.9:1    | 2.4:1    | 1.2:1    |
| Weed precision rate (%)    | 77.0   | 90.1     | 82.4     | 71.6     |
| Crop precision rate (%)    | —      | 28.5     | 33.1     | 38.2     |
| Soil precision rate (%)    | 23.0   | 22.3     | 31.5     | 42.8     |
| Weed-aligned distance (mm) | 7.5    | 10.2     | 15.7     | 21.3     |
| Crop-aligned distance (mm) | —      | 36.1     | 31.8     | 38.5     |
| Spray-to-weed IoU          | 0.53   | 0.16     | 0.12     | 0.08     |
| Spray-to-crop IoU          | —      | 0.11     | 0.09     | 0.07     |

### 3.6 Comparison with Commercial Viability Benchmarks

Commercial viability requires concurrent achievement of three criteria established in the precision spraying literature: weed control efficacy exceeding 90%, matching the standard achieved by conventional broadcast herbicides (Slaughter et al., 2008); crop damage below 10%, representing the economically acceptable yield loss threshold for vegetable production with narrow profit margins (Slaughter et al., 2008); and operating speed of at least 0.67 m/s (1.5 mph), the minimum for field efficiency in specialty crop operations identified by Vijayakumar et al. (2025). All three criteria must be satisfied simultaneously, as a deficiency in any single dimension renders the system unsuitable for commercial production.

The best field performance achieved in this study, at 0.13 m/s (0.3 mph), nonetheless did not meet the three criteria. Weed hit rate reached only 78.3%, falling 11.7 percentage points short of the 90% target. Crop contact rate was 20.1%, exceeding the 10% commercial tolerance by 10.1 percentage points. Operating speed itself fell short by the widest margin: even at its best-performing (and slowest) tested speed, 0.13 m/s, the system operated 80.6% slower than the 0.67 m/s (1.5 mph) minimum required for commercial field efficiency. These gaps are best read diagnostically rather than as a verdict on the approach itself. Detection accuracy remained comparatively stable across the tested range (mAP@50: 94.6% at 0.13 m/s, declining by roughly 24 points to 0.45 m/s), while weed hit rate collapsed more than 43 percentage points over the same interval. That asymmetry is the central finding of this evaluation: the shortfall against commercial thresholds originates downstream of detection, in the tracking-to-actuation stages that convert a detection into a spray event, not in perception itself. Locating the bottleneck there, rather than in the vision model, points to where the next iteration should focus. Velocity estimation uncertainty of 5–10% from wheel-slip-based speed sensing amplifies spray-timing position error 3.1-fold across the tested range (§4.1), a mechanism unrelated to detection accuracy and correctable through RTK-GPS-based velocity estimation and faster valve actuation. Closing the gap to commercial thresholds is hence an actuation- and timing-engineering problem at this point, rather than a detection problem, a distinction that single-speed, detection-only evaluations elsewhere in the literature cannot make.

Linear regression across the three field speeds quantifies degradation rates and enables extrapolation to the commercial threshold. Note that  $R^2$  values approaching 1.0 are mathematically expected when fitting a line to only three data points (**Fig. 7**) and should not be interpreted as strong statistical validation of linearity; these models serve as simple first-order approximations only:

$$mAP@50(v) = 105.0 - 76.8v \quad (R^2=0.998) \quad (9)$$

$$WHR(v) = 95.6 - 135.1v \quad (R^2 = 1.000) \quad (10)$$

$$CCR(v) = 16.8 + 26.8v \quad (R^2=0.997) \quad (11)$$

where  $v$  is platform velocity in m/s and outputs are in percentage points. Extrapolating to 0.67 m/s (1.5 mph) predicts a further severe collapse in performance, with weed hit rate falling to  $\approx 5\%$  (Table 8), rendering the system agriculturally ineffective at commercial operating speed.

**Table 8.** Extrapolated system performance at the commercial threshold speed (0.67 m/s). Extrapolated values assume linear degradation continues beyond the tested range; actual performance may degrade faster due to non-linear motion effects. WHR = Weed Hit Rate; CCR = Crop Contact Rate.

| Metric     | Best tested (0.13 m/s) | Extrapolated (0.67 m/s) | Commercial threshold |
|------------|------------------------|-------------------------|----------------------|
| mAP@50 (%) | 94.6                   | 53.5                    | —                    |
| WHR (%)    | 78.3                   | 5.1                     | >90                  |
| CCR (%)    | 20.1                   | 34.7                    | <10                  |

Across the tested speed range, performance degraded through four compounding patterns. Detection recall declined steadily, falling 30.7 percentage points from the slowest to the fastest speed. Crop contact rates remained elevated at 20–29% across all three speeds, consistently exceeding the 10% commercial tolerance even under the most favorable conditions. The gap between detection and spraying performance widened substantially with speed, growing from 16.3 to 35.1 percentage points as translation-efficiency losses accumulated through the tracking–actuation pipeline. Spatial targeting also deteriorated: aligned distance increased 2.1-fold and spray-to-weed IoU fell by half, indicating that even successfully triggered sprays landed less precisely on their intended targets as speed increased.

## 4. Discussion

### 4.1 Compounding Pipeline Losses, Not Detection Accuracy, Drive Speed-Dependent Degradation

The central finding of this multi-speed field characterization challenges a widely held assumption: that improving detection accuracy is the primary path to commercial viability. Across the tested speed range, detection recall declined 30.7 percentage points while weed hit rate declined 43.3 percentage points, and translation efficiency (the fraction of detected weeds actually sprayed) fell from 84% to 56%, confirming that errors compound multiplicatively through the detection–tracking–actuation pipeline rather than residing in any single stage. Four primary limitations drove this: detection recall degradation under motion, excessive crop contact at all speeds, widening spray timing errors with speed, and degraded spatial targeting.

These findings also challenge the broader literature's reliance on single-speed evaluations, which systematically obscures speed sensitivity and creates an overly optimistic picture of operational readiness; the systematic three-speed design used here instead isolates where in the pipeline performance actually breaks down, providing the evidentiary basis for the prioritized improvement roadmap developed below

rather than a generic call for "better detection". Vijayakumar et al. (2025) determined their reported 0.45–0.89 m/s operating range through a systematic frame-capture analysis across three tested speeds (0.45, 0.89, and 1.34 m/s, each repeated twice), establishing via RTK-GPS-synced frame counts that at least six to seven continuous frames containing the target weed were needed for reliable actuation, a criterion satisfied up to 0.89 m/s but not above 1.12 m/s. Notably, Their lowest tested speed (0.45 m/s) corresponds to this study's highest tested speed, at which our own detection and spraying performance had already degraded substantially; the two systems' validated operating envelopes therefore do not overlap under the conditions tested here. This gap is not evidence that speed tolerance is an arbitrary, non-generalizable property of each system, but reflects a concrete, addressable difference in engineering approach: Vijayakumar et al. 's system combined proportional-valve control with a frame-count-based targeting criterion specifically tuned to sustain reliable actuation at higher speeds, whereas the present system's fixed, non-speed-adaptive latency budget (§2.6.3) and declining frame density (Eq. 3) were not engineered for sustained operation above roughly 0.3 m/s. Closing this gap is therefore an engineering target for the roadmap in §4.2, not evidence that robust, speed-tolerant precision spraying is unattainable.

The speed-dependent degradation reflects compounding physical constraints that persist regardless of algorithm choice. Frame density dropped 71% across the tested range, reducing per-plant detection opportunities from 230 to 67 frames per meter, while motion blur increased pixel displacement 3.4-fold, rendering sub-centimeter morphological features of small weeds indistinguishable from background texture. The differential recall-versus-precision pattern confirms weeds became undetectable rather than misclassified, with precision remaining above 76% even at the fastest speed. These detection losses then propagate and amplify through the actuation pipeline: fixed system latencies totaling ~64 ms produce position errors that scale linearly with speed, shifting spray from target center to edge or beyond, while velocity estimation uncertainty of 5–10% further amplifies positional error 3.1-fold across the tested range. ByteTrack's two-stage association proved effective for handling partial occlusion and confidence fluctuations, but tracking gains were limited because spray timing constraints dominated; weed hit rates consistently fell below corresponding recall values. Together, these mechanisms explain why even a perfected detection model would still face substantial spraying degradation at commercial speeds. The improvement roadmap must prioritize hardware interventions, such as higher-frame imaging, the use of fast-responsive pulse-width modulation (PWM) valves in place of the present valve (~18 ms solenoid response), and more robust platform speed estimates independent of wheel slip.

Multi-season training across five growing seasons addressed spatial generalization: field performance at the slowest speed exceeded the dataset-based baseline, likely reflecting ByteTrack's temporal integration recovering missed single-frame detections, though this advantage disappeared at higher speeds, consistent with dataset diversity addressing covariate shift across field conditions and

morphologies, while motion-induced concept drift, absent from any static training image, remains a separate and unresolved challenge. Controlled LED illumination appears to have contributed to field generalization: the 4.5-point indoor-to-field gap observed here at slow speed is smaller than gaps commonly described for vision systems operating without illumination control, though we were unable to identify a directly comparable benchmarked figure in the literature, and this comparison should be read as a qualitative observation rather than a quantified claim. A comparable dataset-to-field performance gap has not been explicitly quantified in our group's prior work (Deng et al., 2026a, 2026c), so the 20-point gap identified here between dataset-based mAP@50 and video-based field performance at higher speeds represents a new empirical finding for this system lineage. It indicates that static image benchmarks alone are insufficient to anticipate the performance degradation that emerges under continuous field motion. This finding is reinforced by Deng et al. (2026b), who evaluated a higher-precision micro-jet sprayer under comparable field conditions. Although the system achieved a field detection accuracy of 80.0% mAP@50, its crop hit rate of 17.6% still exceeded the 10% commercial tolerance despite finer actuation. This further confirms, using different hardware, that field-based validation is necessary to anticipate real-world degradation, and that crop-adjacent weed targeting remains an open problem requiring continued investigation.

## **4.2 Crop Damage: The Primary Barrier**

Crop contact rates of 20–29% at all tested speeds represent a critical barrier to practical deployment, exceeding the established 10% tolerance threshold (Slaughter et al., 2008) by approximately 2–3×. Three mechanisms explain this elevated crop contact. First, spatial proximity constrained targeting precision: weeds within 2–5 cm of crop plants fell within the positional error envelope of the present solenoid-actuated, 12-nozzle sprayer given the 2.5 cm in-row radish spacing used here. Second, morphological similarity between young radish and weeds (Palmer amaranth) likely contributed to classification errors that the  $\text{IoU} > 0.25$  crop overlap filter could not fully suppress. Third, speed-dependent timing errors increasingly displaced spray toward adjacent crop plants at higher platform speeds. Together, these mechanisms reflect limitations in the spatial and temporal precision of the present actuation pipeline rather than an absolute physical limit on crop-adjacent targeting. A recently reported sprayer using the same 12-nozzle, 1 cm nozzle-pitch geometry as the present system, but with micro-jet rather than solenoid actuation, achieved a lower crop hit rate of 17.6% under comparable field conditions (Deng et al., 2026b), indicating that actuator technology, not nozzle spacing, is a meaningful lever for reducing crop contact. Notably, however, even this higher-precision sprayer remained above the 10% commercial tolerance. This result is consistent with Deng et al. (2026b), who concluded that further reducing crop contact will likely require more robust weed detection and an explicit buffer zone around crops, in addition to actuator improvements. Addressing this barrier in the present system therefore requires a

coordinated hardware-software strategy incorporating explicit weed–crop distance thresholding, faster valve actuators or alternative actuation technologies, and appearance-based association within the existing tracking pipeline (e.g., incorporating visual feature similarity alongside the current IoU-based matching in ByteTrack) to reduce identity switches between morphologically similar species.

These results characterize the present system's crop-adjacent targeting precision rather than offering a general verdict on vision-guided precision spraying relative to other weed control strategies; a full economic and performance comparison against broadcast herbicide, mechanical cultivation, and laser weeding is beyond the scope of the single-platform, single-crop trials reported here. What this study does establish, corroborated by the independent, higher-precision system discussed above, is that spray-timing latency and actuator resolution, not detection accuracy, are the binding constraints on crop-adjacent targeting across at least two independent system generations in this research program, and that closing them is a well-defined engineering target for near-term development, discussed next.

Several limitations constrain the generalizability of these findings and define important directions for future work. Testing was conducted at a single site on one crop species under favorable conditions (flat terrain, minimal wind, overcast sky) with two dominant weed species. These conditions may limit extrapolation to the broader diversity of vegetable production environments. The tested speed range of 0.13–0.45 m/s did not reach the 0.67 m/s commercial threshold, requiring extrapolation that likely underestimates actual degradation given the non-linear acceleration of performance losses observed between 0.27 and 0.45 m/s. The spray-timing model's valve-response and droplet-flight latency terms (§2.6.3) were based on nominal values rather than bench-validated measurements for the hardware used in this study; only the software-processing component was directly measured (Table 5). All three field speed treatments were tested sequentially, single-pass, on the same experimental rows rather than on independent, replicated plots. Although the approximately 3–4 h interval between speeds was not expected to confound ground-truth scoring, dye carryover and row-order effects were not independently verified. The use of water-dye markers provided a proxy for spray deposition rather than herbicide efficacy, leaving actual weed mortality unvalidated.

Further research should therefore prioritize extending speed characterization to and beyond commercial operating thresholds, conducting multi-crop validations, and improving motion-robust perception through video object detection with temporal attention. Integration of RTK-GPS velocity measurements could further improve timing compensation across variable platform speeds. Additional work is still needed to bench-validate valve response and droplet-flight latency, evaluate faster valve actuators or alternative actuation technologies, and conduct herbicide efficacy trials that directly quantify weed mortality. Longer-term research should include multi-site validation, economic comparisons with conventional and alternative weeding methods, adaptive speed control based on real-time weed density,

and continued actuator-technology development focused specifically on overcoming the crop-adjacent contact limitation identified here as the primary practical barrier.

## **5. Conclusion**

This multi-speed field evaluation shows that spraying accuracy degrades faster than detection accuracy as platform speed increases: weed hit rate fell 43.3 percentage points, compared with 30.7 points for detection recall, indicating that downstream tracking and actuation, not perception, are the primary constraint on commercial viability. Crop contact (20–29% across all tested speeds) was the more critical barrier, exceeding the 10% tolerance threshold regardless of operating speed. The system met none of the three commercial-viability criteria ( $>90\%$  weed control,  $<10\%$  crop damage,  $\geq 0.67$  m/s operating speed), even at its best-performing, slowest tested speed. These results point to concrete engineering targets, motion-robust detection, spray-timing compensation, and crop-adjacent actuation precision, rather than a fundamental barrier to vision-guided precision spraying in vegetable production.

## **CRedit authorship contribution statement**

Md Khairul Islam: Conceptualization, Methodology, Software, Investigation, Data curation, Formal analysis, Visualization, Writing – original draft. Boyang Deng: Investigation, Data curation, Methodology. Yuzhen Lu: Conceptualization, Methodology, Supervision, Project administration, Funding acquisition, Writing – review & editing. Daniel Brainard: Resources, Investigation. Ajit Srivastava: Resources, Investigation.

## **Declaration of competing interests**

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## **Data availability**

The annotated dataset (Weed&CropDet2025 component) and trained model weights will be made available upon reasonable request to the corresponding author. The previously published 3SeasonWeedDet10 dataset is available from Deng and Lu (2025). Code for the detection, tracking, and

spray-timing control pipeline is available at <https://github.com/AgFood-Sensing-and-Intelligence/multispeed-precision-sprayer>.

## Supplementary material

- **Supplementary Table S1.** Complete hardware specifications and navigation retrofit cost breakdown for the semi-autonomous precision spraying platform described in §2.1–§2.4.
- **Supplementary Table S2.** Complete training hyperparameters for the YOLOv11 detection model, as referenced in §2.6.1.
- **Supplementary Table S3.** Complete failure mode taxonomy for unsuccessful spray events (missed weeds and crop contacts), as referenced in §2.8.3.
- **Supplementary Video S1.** Field detection, tracking, and live spray actuation recorded during the multi-speed trials (August 29–September 4, 2025), illustrating the targeting behavior underlying the results in Table 7.

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